Project Wonky Sprout: Architectural Specifications for a Neuro-Inclusive Social Construction Ecosystem (The "Constructor's Challenge" Initiative)

1. Executive Summary and Strategic Alignment

The following documentation constitutes a comprehensive architectural framework for "Wonky Sprout OS," a bespoke, sociotechnical operating system designed to facilitate collaborative creative work within a neurodivergent (ND) family unit. This report provides the structural specifications for the "Constructor's Challenge"—a gamified, iterative development cycle focused on the fabrication of collaborative "social helmets." This initiative is not merely a recreational project; it is a structured therapeutic and educational intervention designed to bridge communication gaps, foster technical literacy, and leverage the specific cognitive strengths of the stakeholders: the Lead Engineer (Father/Architect), the Operator (9-year-old son), and the Artisan (6-year-old daughter).

The strategic imperative for this project is derived from the intersection of high-level professional expertise and specific personal necessity. The Lead Engineer, William R. Johnson, possesses a sophisticated background in submarine electrical systems, root cause analysis, and rigorous industrial quality assurance as an Engineering Technician at the Trident Refit Facility (TRF).[1] Concurrently, the family navigates the complexities of neurodivergence, specifically Autism Spectrum Disorder (ASD), which presents unique challenges in nonverbal communication and the alignment of internal intent with external expression.[1]

The "Constructor's Challenge" functions as a dual-purpose vehicle. First, it serves the **Technical Objective** of designing and fabricating Assistive Technology (AT) devices—specifically, interactive helmets—that externalize internal emotional states using electronics, adhering to the Lead Engineer's stated interest in bridging the communication gap.[1] Second, it fulfills the **Developmental Objective** of creating a "scaffolded" social environment where the children engage in co-creation. This utilizes their specific interests

(gaming and art) to participate in a rigorous engineering workflow that mirrors the professional standards of the TRF [1] but is adapted for a domestic, gamified context.

This report moves beyond simple project ideas into a holistic "Operating System" for family interaction. Wonky Sprout OS utilizes the "Constructor's Challenge" as its primary kernel, processing inputs (creativity, sensory needs, technical constraints) and generating outputs (functional AT devices, enhanced social cohesion, technical skill acquisition). The report draws heavily on the principles of Constructionism, the Zone of Proximal Development, and the specific industrial safety cultures of naval engineering to create a robust, failsafe environment for learning and connection.

2. Stakeholder Profile Analysis and Capability Assessment

The success of any complex engineering project relies on a nuanced understanding of the workforce. In this context, the "workforce" is the family unit. By analyzing the available data on the Lead Engineer and inferring the profiles of the children based on typical ND presentations and the provided descriptors, we can tailor the Wonky Sprout OS to maximize engagement and minimize friction.

2.1 The Lead Engineer (The Architect)

The project is anchored by the father, identified as the "Architect." His profile serves as the bedrock for the system's technical rigor.

- **Professional Identity:** William R. Johnson, Engineering Technician, GS-0802-12.[1]
- **Core Technical Competencies:**
  - **High-Level Electromechanical Systems:** His extensive experience with 500kW motor generators, Voltage Frequency Regulators (VFRs), and complex submarine propulsion systems [1] indicates a capability far beyond typical hobbyist electronics. This allows for the design of highly sophisticated internal electronics for the helmets, potentially incorporating advanced logic or power management systems.
  - **Procedural Rigor:** The Architect's professional life is governed by Maintenance Standards (MS), Industrial Process Instructions (IPI), and Quality Assurance (QA) forms.[1] This suggests a cognitive need for structure; the home project will likely fail if it is chaotic. It requires the same "instruction card" logic used at Kings Bay.[1]

- **Training & Mentorship:** He has documented success as "Production Employee of the Year" specifically for leading training efforts and passing critical knowledge to junior electricians.[1] This is the critical "unlock" for the project: The father is not just a builder; he is a professional trainer. The project should be viewed as an apprenticeship program.
- **Neurocognitive Profile:** Late-diagnosed ASD. Identified challenges include nonverbal communication and facial expressions not matching feelings.[1] Strengths include an intense, preoccupying interest in how things work ("taking things apart") and a drive to create custom solutions.[1]

- **Role in Wonky Sprout OS:** The Architect provides the "skeleton" of the experience. He designs the safety protocols, maintains the hardware infrastructure (3D printers, soldering stations), and creates the "Quest Logs" (Work Packages).

2.2 The Operator (9-Year-Old Son)

- **Profile:** "Gamer."
- **Inferred Neurocognitive Strengths:**
  - **Systemizing:** Gamers often excel at understanding complex rule sets, optimizing variables (min-maxing), and following logic trees. This translates well to the logic required for the "Haystack" inventory system [1] and coding logic.
  - **Persistence:** Willingness to repeat tasks (grinding) if the reward structure (XP, loot) is clear.
  - **Spatial Reasoning:** Navigating 3D game environments translates directly to CAD (Computer-Aided Design) literacy.
- **Potential Friction Points:** Low tolerance for boredom, frustration with open-ended tasks without clear win conditions, potential sensory sensitivities.
- **Role in Wonky Sprout OS:** "Systems Integration & CAD Lead." His gaming interest is the bridge to CAD software (Tinkercad or Blender) and coding. He is motivated by the "gamification" layer of the OS.

2.3 The Artisan (6-Year-Old Daughter)

- **Profile:** "Artist."
- **Inferred Neurocognitive Strengths:**
  - **Visual-Spatial Processing:** Acute awareness of color, texture, and form.
  - **Divergent Thinking:** Ability to generate novel aesthetic ideas that strict engineering logic might miss.

    - **Fine Motor Skills:** (Developmentally appropriate) Drawing, painting, manipulating tactile materials.
- **Potential Friction Points:** May struggle with the abstract nature of code or the rigidity of engineering specs. Needs tangible, immediate feedback.
- **Role in Wonky Sprout OS:** "Aesthetics & User Experience (UX) Lead." She defines the "look and feel" of the helmets, translates emotions into color codes, and handles the physical finishing of the 3D printed parts.

3. The Wonky Sprout OS Architecture: Theory and Framework

Wonky Sprout OS is not software installed on a computer; it is the *methodology* of collaboration. It is a "social operating system" that governs how the family interacts during the build process. It borrows heavily from the "Formal Work Packages" (FWP) and "Controlled Work Packages" (CWP) used by the Lead Engineer at the Trident Refit Facility [1], but wraps them in a fantasy/sci-fi narrative suitable for children.

3.1 Core Kernel: The "Constructor's Challenge" Loop

The "Constructor's Challenge" is the primary process loop of the OS. It breaks the massive task of "building a helmet" into bite-sized, gamified missions. This prevents the "overwhelm" often associated with ASD projects.

The Loop Structure:

1. **Briefing (The Download):** The Architect presents the mission parameters (e.g., "We need to design a cooling vent for the helmet"). This mimics the "Pre-Job Brief" the Architect delivers to journeymen.[1]
2. **Drafting (The Blueprint):** The Artisan sketches the concept; The Operator models it in CAD.
3. **Fabrication (The Forge):** The Architect oversees the 3D printing and hardware assembly, involving the children in safety checks and "receipt inspections".[1]
4. **Integration (The Link):** Electronics are installed and coded.
5. **Debrief (The After-Action Report):** A review of what worked, what failed, and the awarding of XP (Experience Points).

3.2 The Environmental Layer: The "Safe Harbor" Makerspace

Given the sensory sensitivities associated with ASD [1], the physical environment must be carefully managed. The workshop is not just a room; it is a "Clean Room" environment, similar to the specialized shops at TRF.[1]

Zone Designation	Activity Profile	Sensory Control	Toolset
Zone Red (The Forge)	Fabrication, Machining	High Noise, High Temp. Ear Defenders required.	3D Printers, Dremels, Power Supplies.
Zone Blue (The Lab)	Electronics, Soldering	Low Noise, Fume Extraction. Bright task lighting.	Soldering Station, Multimeters, Breadboards.
Zone Green (The Studio)	Design, Finishing	Quiet, Natural Light. Tactile materials.	Paints, Sandpaper, Sketchpads, Tablets.
Zone Zero (Safe Harbor)	Regulation, Rest	Silent, Dim. Beanbags, Weighted blankets.	None. This is a "No Work" zone.

3.3 The Theoretical Underpinning: Constructionism and the ND Mind

To fully understand *why* the Constructor's Challenge is the optimal approach for this specific user profile, we must explore the pedagogical theories of Seymour Papert's **Constructionism**. Papert argued that learning happens most effectively when people are actively engaged in making tangible objects in the real world. For the Neurodivergent mind, this is particularly potent.

The Architect's resume indicates he is "highly motivated" and "self-driven".[1] This aligns with

the "Interest-Based Nervous System" common in ND individuals, where motivation is regulated by interest, challenge, novelty, and urgency rather than importance or rewards. Wonky Sprout OS is designed to hack this system:

- **Interest:** Leveraging the Son's gaming and Daughter's art.
- **Challenge:** The "Level Up" mechanics of the XP system.
- **Novelty:** The "Loot Drops" of new components.
- **Urgency:** "The printer is running; we have 10 minutes to splice this wire!"

4. Phase 1: The Blueprint (Concept & Design)

The first phase of the Constructor's Challenge focuses on collaborative design. The goal here is to integrate the Artisan's aesthetic vision with the Operator's structural logic, facilitated by the Architect's engineering constraints.

4.1 Mission 1: The "Skin" Design (Involving the Artisan)

The 6-year-old artist is the Creative Director for the helmet's exterior.

- **The Challenge:** "The helmet needs to look like it belongs to a Space Explorer/Deep Sea Diver."
- **The Workflow:**
  1. The Architect provides a printed template of a basic helmet shape (a "wireframe" on paper).
  2. The Artisan uses markers, clay, or collage to design the "skin."
  3. **Insight:** This validates the daughter's contribution immediately. She is not "helping"; she is *defining* the project.
- **Technical Translation:** The Architect teaches the children how 2D drawings become 3D objects. This mirrors the process of interpreting "manufacturer's drawings" [1] but in reverse—creating the specs rather than following them.

4.2 Mission 2: The Digital Twin (Involving the Operator)

The 9-year-old gamer takes the Artisan's design and digitizes it.

- **The Challenge:** "Translate the Artisan's drawing into a 3D model we can print."
- **The Toolset:** Tinkercad (entry-level) or blocked-out shapes in Minecraft (if that is his gaming preference), which can be exported to.STL files.
- **The "Gamification" Hook:** Frame this as "modding" the real world. Just as he might mod a character in a game, he is modding his sister's helmet.
- **The Skill Tree:** The Operator earns XP in "Spatial Geometry" and "CAD Logic." The Architect provides "cheats" (advanced CAD shortcuts or pre-made assets) when frustration mounts, acting as the "Senior Tech" resolving a blockage.[1]

4.3 The Architect's Oversight: Feasibility Analysis

While the children design, the Lead Engineer performs the "Engineering Technician" role.[1]

- **Review:** Does the design fit the print bed? Are the overhangs printable?
- **Scaffolding:** Instead of saying "no," the Architect poses engineering challenges. "If we put the spike there, the printer nozzle will hit it. How do we move it?" This introduces "Root Cause Analysis" thinking [1] at an early age.

5. Phase 2: The Foundry (Fabrication & 3D Printing)

This phase leverages the Lead Engineer's deep expertise in 3D printing and materials.[1] It is the physical manifestation of the digital dream.

5.1 The "Material Request Form" Protocol (Logistics Training)

To teach the Operator the value of logistics (a key part of Will's job interacting with Supply Code 500 and systems like One Touch [1]), the child must submit a "Material Request."

- **The Activity:** The Operator checks the filament inventory.
- **The Logic:** "We need 200g of Blue PLA. Do we have it?"
- **The Lesson:** Inventory control and resource management. The Architect can set up a simplified "Haystack" database (perhaps a spreadsheet or a physical card catalog) where the Operator looks up filament by "Part Number." This connects the child's play directly

to the father's professional expertise in logistics management.[1]

5.2 Collaborative Machine Operation

The Architect knows that "a happy employee is a safe employee".[1]

- **For the Operator:** He is responsible for "Pre-Flight Checks" (bed leveling check, nozzle temp check). This mirrors the "Receipt Inspections" Will performed on motors.[1]
- **For the Artisan:** She selects the filament color and watches the "first layer" specifically to report any "adhesion anomalies" (bubbles or peeling).
- **Safety Integration:** This is the ideal time to introduce the concept of PPE (Personal Protective Equipment).[1] The "Constructor's Uniform" (lab coats, goggles) makes safety a role-playing element rather than a chore.

5.3 Post-Processing: The Artisan's Domain

Once the print is done, the raw part is rough.

- **The Task:** Sanding, priming, and painting.
- **The Fit:** This plays perfectly to the Artisan's fine motor skills and artistic interest.
- **The "QA" Check:** The Architect introduces "QA Forms".[1] The Artisan must inspect the part for "defects" (layer shifts, stringing) and mark them with a crayon on a printed sheet. This teaches the concept of "Objective Quality Evidence" (OQE) [1] in a playful way.

6. Phase 3: The Circuitry (The "Social Engine")

This is the most critical phase for the Lead Engineer's personal goals: building an AT solution for communication.[1] The helmet is not just a shell; it is a machine that helps the wearer express emotion.

6.1 The "Empathy Circuit" Concept

The core feature of the helmet is an external display (LED Matrix) that reflects the wearer's internal state. This directly addresses Will's challenge of facial expressions not matching feelings.[1]

- **The Mechanism:** An LED matrix (8x8 or 16x16) mounted on the forehead or visor.
- **The Control:** A control pad on the wrist (gauntlet) with buttons corresponding to specific emotions (Happy, Overwhelmed, Focus Mode, Do Not Disturb).

6.2 Wiring the Nervous System (The Operator's Challenge)

The 9-year-old helps build the "nerves" of the helmet.

- **Task:** Breadboarding the circuit. Connecting the Microcontroller (Arduino/ESP32) to the LED matrix.
- **The Skill:** Logic flow. "If Button A is pressed, THEN Light B turns Green."
- **The Architect's Role:** Will uses his expertise in "electrical schematics" [1] to draw simplified wiring diagrams for the son to follow. He teaches the safe use of the soldering iron (under strict supervision), treating it with the reverence of high-voltage test equipment.[1] The father's experience with "encapsulation" [1] can be taught here—using heat shrink tubing to protect the wires, explaining that this is how they waterproof submarine cables.

6.3 Coding the "Social OS" (The Shared Challenge)

Will is "teaching himself to code" and is interested in AI.[1] This is a journey he shares with the Operator.

- **The Language:** Block-based coding (MakeCode or Scratch for Arduino) is the bridge.
- **The Logic:** "When I feel [Emotion], I want the helmet to show [Icon]."
- **Artistic Input:** The Artisan draws the icons (a pixelated smiley face, a storm cloud for 'overwhelmed'). The Operator codes the pixel map.
- **ND Inclusion:** This allows the children to define their *own* emotional language. Perhaps "overwhelmed" isn't a sad face; maybe it's a spinning vortex. This empowers them to communicate on their terms.

7. The "Gamification" Layer: XP and Skill Trees

To keep the children engaged through the tedious parts of engineering (sanding, debugging code), Wonky Sprout OS implements a strict Gamification Layer.

7.1 The Rank System

Modeled after the naval ranks or a sci-fi crew, but adapted for the family.

- **Level 1: Scavenger.** (Can gather parts, perform simple inspections).
- **Level 2: Tinkerer.** (Can operate the 3D printer menu, basic sanding).
- **Level 3: Fabricator.** (Can solder with supervision, edit CAD files).
- **Level 4: Technomancer.** (Can write code loops, design complex assemblies).

7.2 The Quest Board

A physical whiteboard in the workshop listing "Active Work Packages."

- *Quest:* "The Filament Drought."
  - *Objective:* Swap the filament spool and verify nozzle flow.
  - *Reward:* +50 XP, "Master of Materials" sticker.
- *Quest:* "The Great Smoothness."
  - *Objective:* Sand the helmet dome until it feels like glass.
  - *Reward:* +100 XP, Choice of dinner/movie.

7.3 "Loot" Drops

Completing major milestones (e.g., "The Electronics Integration") unlocks "Loot Boxes." These contain not candy, but "Rare Components"—special LEDs, glow-in-the-dark filament, or new tools for their personal kits. This reinforces the "interest-based nervous system" typical of ADHD/ASD, where novelty and interest drive motivation.

8. Neuro-Inclusive Considerations and "Safety Flashes"

The Lead Engineer is responsible for "developing positive safety and health attitudes".[1] In a home with ND children, "safety" extends to emotional and sensory safety.

8.1 The "Meltdown Protocol" (Emergency Stop)

Just as industrial machinery has an E-Stop, the Constructor's Challenge has a social E-Stop.

- **The Signal:** A specific hand gesture or a red button on the workbench.
- **The Action:** If anyone (Architect included) gets overwhelmed, the "Safety Flash" is triggered.[1] Work stops immediately. No questions asked.
- **The Recovery:** The team moves to the "Quiet Zone" for regulation. This teaches the children that their sensory needs are valid and respected, mirroring the safety culture Will enforces at work where "A happy employee is a safe employee".[1]

8.2 Parallel Play and The "Safety Flash" Mechanism

ND children often prefer "parallel play" (working near others but on their own tasks). The workshop layout should support this. The Artisan can be painting at her station while the Operator codes at his, and the Architect manages the printer. They are "together" in the shared goal (The Helmet) but autonomous in their actions.

The resume mentions utilizing "Safety Flashes" and "Training Bulletins".[1] In the Wonky Sprout context, these are not bureaucratic memos but "Status Updates."

- **Example Flash:** "ATTENTION: The soldering iron is now HOT. Zone Blue requires Safety Glasses."
- **Pedagogy:** This formalizes the transition between safe and hazardous states, helping the children switch cognitive gears.

9. Deep Dive: The Safety & Quality Assurance (QA) Sub-System

Given Will's background at TRF, safety and QA are not afterthoughts; they are the foundation. This section details how to translate industrial safety standards [1] into a domestic setting.

9.1 The "Wonky Sprout" Safety Manual (Ref: TRF Inst 5100.1E)

The Architect should draft a physical safety manual. This is not a boring document; it is the "Rulebook of the Forge." It adapts the high-level safety culture of submarine refit [1] for family use.

Table 1: Hazard Control Hierarchy for the Home Workshop

Hierarchy Level	Industrial Application	Wonky Sprout Application
Elimination	Remove hazard completely	Use non-toxic PLA instead of ABS (no fumes). Eliminate lead solder.
Substitution	Replace with safer alternative	Use low-temp soldering irons or conductive thread for the 6yo.
Engineering Controls	Physical barriers/guards	Enclosure for the 3D printer (keeps fingers out). Fume fan.
Administrative Controls	Training & Procedures	"Tag-out" system: The 3D printer cannot be started without the "Key."
PPE	Goggles, Gloves, Suits	"Constructor's Goggles"

		(Cool styling), Lab Coats, Ear Defenders.

9.2 The "Tag-Out" Simulation

At TRF, a "Tag-Out" prevents machinery from being energized while worked on. This is a critical life-safety procedure for an electrician.[1]

- **The Game:** Create physical tags (Red Tags for "Danger", Yellow for "Caution").
- **The Lesson:** If the Operator is changing the printer nozzle, he hangs a RED TAG on the power switch. The Architect *must* respect this tag. If the Architect tries to turn it on, the Operator gets to "fine" him (loss of XP). This empowers the child to police the safety of the environment, enforcing mindfulness. It teaches the children that safety rules apply to *everyone*, even the boss.

9.3 Objective Quality Evidence (OQE) and The QA Form

Will's resume mentions capturing "Objective Quality Evidence" to certify maintenance.[1] This is the process of proving that work was done correctly.

- **The "Certificate of Seaworthiness":** Before a helmet can be worn in a "Live Fire" social scenario, it must pass QA.
- **The Inspection Checklist (for the 9yo):**
  - [ ] No sharp edges (Tactile check).
  - [ ] Battery voltage > 3.7V (Multimeter check - involves using tools).
  - [ ] Strap tension secure (Mechanical check).
  - [ ] LED "Happy" sequence triggers correctly (Functional check).
- **Signature Required:** The Architect signs off as "Quality Assurance Director." The Operator signs as "Lead Technician." This ritual adds immense weight and pride to the accomplishment.

10. Technical Specification: The "Social Helmet" Hardware

This section provides the "Build of Materials" (BOM) and technical rationale for the helmet, catering to Will's expertise in electronics and 3D printing [1] and his familiarity with motor generator sets and regulators.[1]

10.1 Chassis Fabrication (3D Printing)

- **Material:** PETG (Polyethylene Terephthalate Glycol) is recommended over PLA for the structural parts due to its flexibility and durability (kids drop things).
- **Filament Choice for Sensory Needs:** "Matte" PLA hides layer lines and feels soft/chalky (good for some sensory profiles). "Silk" PLA is shiny and smooth. The Artisan should choose the texture.
- **Design Modularization:**
  - *Core:* A headband/halo structure (prints flat, easy fit).
  - *Modules:* Snap-on "ears," "visors," or "vents." This allows for iterative upgrading without re-printing the whole helmet.

10.2 The "Comm-Link" Electronics Package

- **Microcontroller:** ESP32-WROOM-32.
  - *Why:* Dual core (handles WiFi and LED drive smoothly). Deep sleep modes to save battery. The "AI" interest [1] suggests Will might eventually want to integrate voice recognition or machine learning; the ESP32 is capable of running lightweight TensorFlow Lite models.
  - *Coding:* Using C++ (Arduino IDE) allows Will to teach the Operator "real" coding concepts (loops, variables, boolean logic).[1]
- **Display:** WS2812B (Neopixel) 8x8 Flexible Matrix.
  - *Why:* Flexible enough to curve around a helmet forehead. Addressable RGB allows for color-coding emotions (Red = Stop, Green = Talk, Blue = Sad).
- **Sensors (Advanced Challenge):**
  - *Pulse Sensor:* Can the helmet detect when the Operator's heart rate spikes (anxiety/excitement) and auto-change the lights to "Pulsing Yellow"? This introduces "Bio-feedback" concepts.
  - *Microphone:* A sound sensor could visualize noise levels on the display. If the room is "Too Loud" for the ND sensory profile, the helmet creates a visual "EQ Bar" that hits red, signaling everyone to lower their voices. This is *functional AT* for sensory

regulation.

10.3 Power Management and VFR Metaphor

- **Battery:** 18650 Li-Ion Cell with a TP4056 charging module.
- **Safety Housing:** The battery must be in a printed "bunker" on the back of the helmet (acts as a counterweight).
- **The VFR Lesson:** Will works with Voltage Frequency Regulators.[1] He can explain the battery management system (BMS) as the helmet's VFR. "Just like the submarine needs steady power, the helmet needs steady voltage. If the voltage drops, the 'brains' (ESP32) get confused." This anchors the new knowledge in his existing professional schema.

11. Leveraging External Ecosystems (The Tools for Life Connection)

The Architect has already initiated contact with the external ecosystem.[1] Wonky Sprout OS should not exist in a vacuum; it should interface with the broader AT community to ensure sustainability and community engagement.

11.1 The "Makers Making Change" Integration

The "Constructor's Challenge" can eventually graduate from building for *self* to building for *others*. Will has been directed to the Makers Making Change website.[1]

- **The Project:** Once the family masters their own helmets, they can download open-source AT designs from Makers Making Change (e.g., bottle openers, key turners).
- **The Benefit:** This teaches the children empathy and the social value of their technical skills. It transforms the "Special Interest" into a "Social Contribution."

11.2 The TOM Global (Tikkum Olam Makers) Connection

Will has been directed to the TOM Global chapter at Georgia Tech.[1]

- **Future Vision:** The family could participate in a "Makeathon." This would be the ultimate "Raid Boss" for the Constructor's Challenge team—taking their skills out of the home and into a community event.
- **Volunteerism:** Will offered to volunteer his skills.[1] Involving the children in this (even passively, by helping pack the printed parts) models civic engagement. It shows them that their father's skills are valued by the community, reinforcing family pride.

12. The "Constructor's Challenge" Curriculum: A 12-Week Roadmap

To provide structure, the project should be broken into a "Semester" or "Season." This prevents the project from becoming an infinite, unfinished task.

Weeks 1-2: The Initiation (Infrastructure)

- Setup the Makerspace zones.
- Design the "Family Logos" (Artisan lead).
- 3D Print the "Initiation Badges."
- Safety Training & "Tag-Out" drills.

Weeks 3-4: The Prototype (Cardboard & Code)

- Build mock-ups using cardboard (rapid prototyping).
- Learn basic "Blink" sketch on Arduino.
- Artisan draws the "Emoticon Library" on graph paper.

Weeks 5-8: The Heavy Lifting (CAD & Fabrication)

- Operator models the helmet attachments using Tinkercad.
- Architect manages the "Long Prints" (overnight jobs).
- Post-processing parties (Sanding & Painting).
- Material Request Forms used for all filament changes.

Weeks 9-10: The Integration (Wiring)

- Soldering the harness (Zone Blue).
- Installing the matrix.
- "Smoke Test" (First power up).

Weeks 11-12: The Deployment (Optimization)

- Field testing the helmets in the house.
- Refining the code (fixing bugs).
- Documentation (writing the "User Manual").
- **Graduation Ceremony:** A family dinner where they present their helmets and receive their "Certificates of Competency."

13. Addressing the User's "Pain Points"

13.1 "Facial Expressions Do Not Match Feelings"

[1]

This is the central problem Will wants to solve. The helmet is the solution.

- **The "Truth Mode":** The helmet can be programmed to have a "Default State" that reflects the user's *true* intent. Even if Will's face looks flat or angry due to concentration, the helmet can display a "Thinking Gear" animation.
- **Externalizing the Internal:** By explicitly coding these states ("When I am focusing, I am not angry"), Will is also teaching his children to interpret him. "Don't look at Dad's face; look at his helmet status." This reduces anxiety for everyone.

13.2 "Involving the Children" (The Collaborative Aspect)

- **The Gamer Son:** The project meets him where he is. It uses the language of games (XP, Loot, Mods) to teach the skills of engineering. It validates his gaming hobby as "Simulation Training" for the real world.
- **The Artist Daughter:** She is often excluded from "hard engineering." Wonky Sprout OS centers her contribution. The helmet is useless without her *visual language* (the icons/colors). She provides the UI/UX (User Interface/User Experience).

14. Conclusion: The "Ready for Issue" State

In the language of the Trident Refit Facility, the goal is to get the equipment "Ready for Issue" (RFI).[1] For Project Wonky Sprout, RFI status is achieved not when the helmets are perfect, but when the *process* of building them becomes a self-sustaining engine of connection.

The "Constructor's Challenge" transforms the home into a laboratory of learning. It validates the Architect's professional identity, utilizes the Operator's gaming drive, and celebrates the Artisan's creativity. It creates a shared language—spoken in code, CAD lines, and LED pulses—that transcends the neurodivergent communication barriers. By applying the rigorous standards of submarine maintenance to the playful world of maker-culture, Will Johnson can build the ultimate Assistive Technology: a stronger, more connected family unit.

End of Report Specifications
Status: Approved for Implementation
Authored by: Senior Systems Engineer (Persona)
Reference Documents: 1

Works cited

1. Gmail - Tools for Life information request.pdf